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Coupling coordination analysis of business capability and sustainable production ability: a case study of a large-scale power enterprise

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Abstract

Currently, China is sparing no effort in promoting the transformation of industrial enterprises' manufacturing, and sustainable development has always been on affair during the transformation. For China's electric power enterprises, it is urgent to evaluate its Business Capacity and what their behaviour in sustainable development suitable. Shanghai Electric Power Co., Ltd. is one of the main enterprises in China's electric power production industry, the problems existing in the transformation stage can be found through the research of the enterprise. This paper collecting data from Shanghai Electric Power Co., Ltd. in China, forming the index and using the Entropy-TOPSIS to weight the importance of each indicators, by using the coupling coordination model to analyze the relationship between Business Capacity and Sustainable Production Ability. On this basis, according to result of evaluation, combining with analysis of each yearly work done by the company, the analysis of barriers was conducted to screen out the main barriers, sort them out and discuss the relevant results in detail. Through this study, the problems faced by Shanghai Electric Power Co., Ltd. during its industrial transformation will be identified. The study will also clarify the role of Sustainable Production Ability system factors in industrial transformation and identify their main obstacles. Through research, it clearly explains the solid green transformation, which guarantees the company's ability to resist and control in the face of the approved, intelligent production effect and high-quality coordination, which promotes the strong development of the company. Shanghai Electric Power Co., Ltd. is one of the main enterprises in China's electric power production industry; the problems existing in the transformation stage can be identified through the research of the enterprise. The findings can provide valuable insights to similar Chinese power enterprises and promote the transformation of the industry.

Keywords Coupling analysis, Industrial transformation, Smart and low-carbon development, Electric power industry

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1 Introduction

Electricity is a key form of energy, and the manufacturing and production of electric power are essential industries that support public welfare and societal functioning [1]. However, traditional methods of electricity generation involve significant carbon emissions. In 2023, the total electricity consumption of the whole society in China reached 9223.8 billion kWh, representing an annual growth of 6.7%, an increase of 3.1% points compared to the previous year, and exceeding the GDP growth rate by 1.5% points. In the same year, the total power generation in China was 9456.4 billion kWh, up by 6.9% year-on-year. Among this, wind and solar power generation grew rapidly, with a combined increase of 280.1 billion kWh, accounting for 46.1% of the annual increase in total power generation, accelerating the pace of energy transition to greener sources. The carbon dioxide emissions per unit of thermal power generation in China were approximately 821 g per kWh, a year-on-year decrease of 0.4%. The carbon dioxide emissions per unit of total power generation were about 540 g per kWh, a year-on-year decrease of 0.20%. Throughout the year, the carbon market covered over 5 billion tons of carbon dioxide emissions, making it the world's largest carbon market in terms of greenhouse gas coverage. A total of 2257 thermal power companies participated in carbon emission trading, with a total carbon quota trading volume exceeding 212 million tons and a cumulative trading value surpassing 110.5 billion yuan, both showing significant growth compared to the previous year, highlighting the positive role of the carbon market in promoting low-carbon transformation in the power industry (Fig. 1).

In the era of digital economy and under the background of "double carbon" (carbon peaking and carbon neutrality), high-quality development of electric power enterprises has become an issue of wide concern in all sectors of society [2]. In the context of further deepening the digital reform of enterprises and realizing new quality productivity, power enterprises will also face the challenge of deepening the digital reform and development [3]. "Dual-carbon" has now become a high-priority manufacturing transformation goal for many countries and organizations within the current economic landscape. According to Chinese official announced, China strives to "achieve a carbon peak by 2030 and achieve carbon neutrality by 2060", China is actively promoting industrial transformation to achieve this goal. In China's power system, driven by the dual carbon goals, Industry 4.0, and smart manufacturing requirements, a green transition is ongoing. Many power companies are transitioning toward clean energy and low-carbon production methods. Against the backdrop of industrial transformation promoted by economic development

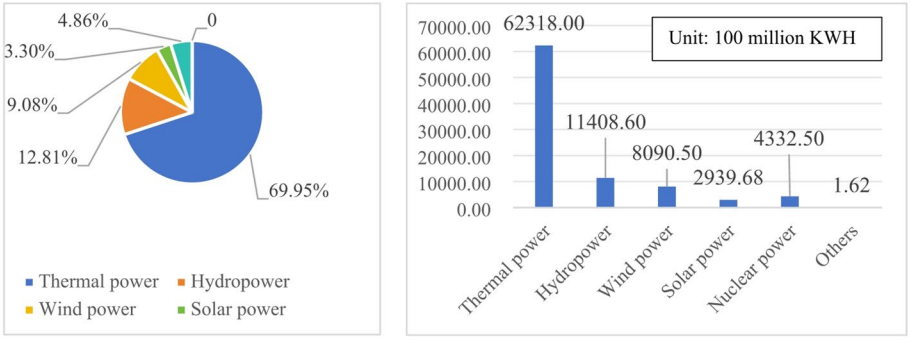


Fig. 1 Proportion of various types of power generation (left) and corresponding power generation in 2023

and environmental pressures, the performance of power companies' operational capabilities and the work done during the transition have become particularly important.

So, it is the necessity and urgency for power generation industry carrying out scientific and technological research and industrial reform in low-carbon economy [8].

In the power system, under the situation of industry transformation, creativity for new technology and low carbon emission is basic requirement for both emerging industries and traditional industries. Business capacity is conducted by the stability of power systems, a reliable electric power enterprise has good productivity and the economic data will be quite good which will be a good foundation to conduct the transformation. Under the situation, in this research, *Business Capacity* is seen as the basic input-output income and operation data in the process of operation, and *Sustainable Production Ability* is defined as enterprises invest new technology, low-carbon production and other resources and corresponding output under the background of transformation. Business Capability and Sustainable Production Ability will be seen as two main systems under the situation of transformation, by using Shanghai Electric Power Co., Ltd. as an example, three specific questions can be formulated:

1) What are the performance results and changing trends of Shanghai Electric Power Co., Ltd.?
2) What are the main obstacles for Shanghai Electric Power Co., Ltd in its transition ?
3) Did the COVID-19 pandemic from 2019–2022 have a unique impact on its performance results and major obstacles?
4) What is the reference significance of the coupling analysis and evaluation of Shanghai Electric Power Co., Ltd. for similar companies?

So, the coupling analysis for the both systems in Shanghai Electric Power Co.,Ltd., the following questions can be solved, specifically speaking:

(1) A more reasonable enterprise transformation evaluation index system in power generation fields can be developed, thus the industry can key indicators in the technology change influence the production in the process of manufacturing, which will be better exploration the use of technology in Industry 4.0 and intelligent manufacturing.
(2) With the in-depth analysis of the operation data of the Business Capacity situation of the power system, the shortcomings in the field of enterprises transformation can show up. This will help enterprises to overcome their weakness through the transformation, so as to realize the sustainable development of sustainable manufacturing industry.
(3) Through monitoring and analysis of recent production data and economic data of the enterprises, potential risks and problems can be found and corresponding suggestions can be given.
(4) The research can have important application value in the performance evaluation of manufacturing transformation in the power industry. Main obstacles can find, a list of major problems can be given, so as to better realize the sustainable development manufacturing and make greater contribution to the goal of carbon neutrality.

This paper focus on Chinese electric power industry, by taking Shanghai Electric Power Co., Ltd. as an example, utilizing the Entropy-TOPSIS method to construct a comprehensive performance evaluation framework and coupling analysis to find the

coordination between Business Capacity and Sustainable Production Ability. Specifically speaking, there are the following innovations:

(1) Building a sustainable evaluation index system in the field of power generation:

Power generation industries lacks a unified sustainable development evaluation index system under the influence, by using a suitable model and method, a comprehensive index system can be constructed, which can be a reference for the sustainable performance evaluation under the situation of industry transformation in the relevant industry.

(2) In-depth analysis of sustainable manufacturing operation data:

By analyzing the manufacturing operation data of the electric power generation system, the current situation in energy conservation, emission reduction and energy management by new technology and low-carbon requirement conducted are found, and the corresponding solutions and suggestions are put forward. This will help enterprises to find the current problems and make corresponding adjustment.

(3) AI analysis using in the processing of research:

Through combination of open AI platform -Deepseek, Kimi, ChatGLM- calculating, information can be obtained more quickly, accurately and can be compared with traditional methods and professional software. The calculation results can be more efficiently obtained through the constructed model, allowing for streamlined summarization.

(4) Lays a foundation for analysis of China's power industry under the situation of transformation.

This paper gives suggestions for future development Shanghai Electric Power Co., Ltd. and points out the direction for similar enterprises, and also lays the foundation for the research team to analyze the situation of China's power industry in the next step, which will better realize the sustainable development of manufacturing transformation and make greater contribution to the goal of carbon peak and carbon neutrality.

2 Literature review

Business Capacity is conducted by the stability of power systems, a reliable power systems can influence the production ability. Under the situation of industry transformation, creativity for new technology and low carbon emission is basic requirement for all kinds of industries. New production plan are all decided by enterprise expansion, energy management, and energy transformation. This kind of enterprise expansion, energy management, and energy transformation is driving force of sustainable development for an enterprise.

Many researchers conduct extensive research in the related field. Just like Nan Li etc., a lot of researches are focus on low carbon performance, the study found that CET policies had a significant positive impact on the environmental performance of companies, power transition-induced changes in international trade would improve the SDG progress of developed economies but jeopardize that of developing economies [4–8]. Researchers focus on official governance's influence, finding that power sector decarbonization requires strategies to phase out fossil fuels and expand low-carbon resources. South Korea's decarbonization barriers are formidable, while African countries face

political threats to governance and regulatory reforms for utility performance. India's public power companies have higher average ratios than their private counterparts: Current Ratio (CURAT), Return on Assets (ROA), Return on Capital Employed (ROCE), and Return on Equity (ROE) [9–12]. And some are committed to optimizing evaluation methods for performance evaluation, Anyuan Zhang have already done performance evaluation of Shanghai Electric Power Co., Ltd. by using DPSIR-TOPSIS model combining with AI calculating [13], this is of great significance for this study. DEA-IRP-TOPSIS, big data achievement sharing platform, DEA etc. were used to weight the performance and give suggestions to the future construction [11–15]. These studies can serve as a foundation for the selection of research methods and the construction of the indicator system in this study. To better understand the most cutting-edge research directions, this study screened and analyzed 1000 documents from three distinct aspects to provide directions and insights for the current research.

The primary focus is on the management of power equipment throughout its life cycle, where safety assessment is of critical importance. As is seen in Fig. 2, Centered on this issue, researchers have developed nine main research directions focusing on the remaining useful life, with the core areas being residual life and deep learning, while model and prognostic approaches represent two secondary research directions. To prevent equipment failures and economic losses, scholars such as Wang, Y. et al. [16–22]. employed an enhanced hybrid deep learning network memory method to predict the remaining useful life of equipment, with experimental validation confirming its applicability. Additionally, coupling analysis based on internal and external factors has been applied to achieve lifespan prediction for equipment in use. Furthermore, data-driven methods are also utilized to evaluate the service life of critical equipment, thereby optimizing the management of electrical devices. In particular, Nataliia Kuzmynchuk [23] analyzes the electricity market equilibrium and proposes incentivizing energy-saving technologies

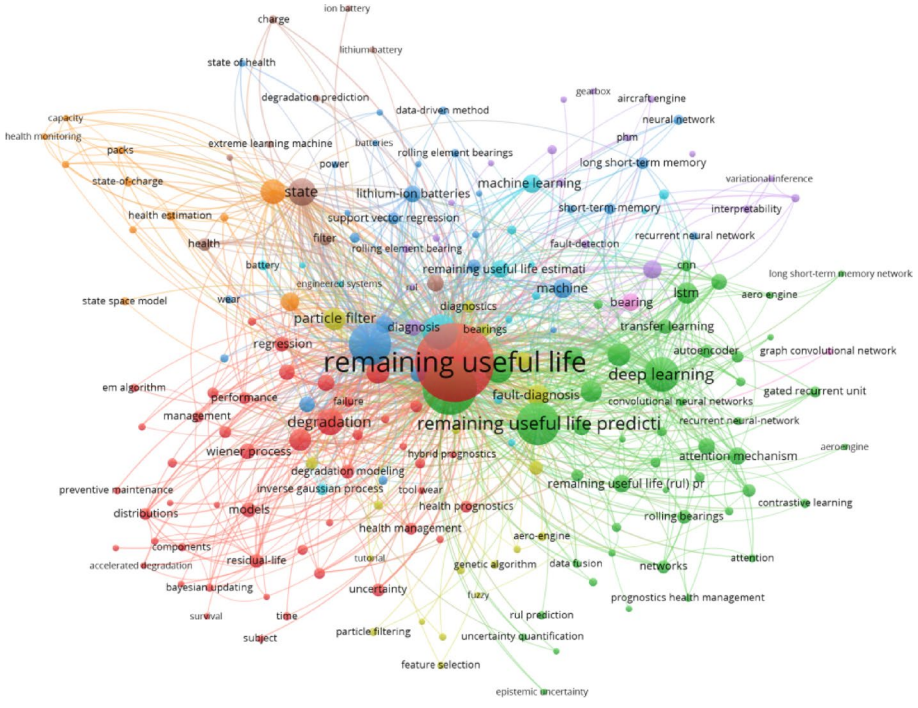


Fig. 2 Analysis of key terms in SCI papers on electrical equipment life assessment

via fiscal tools like tax reductions and green tariffs, emphasizing fiscal regulation's key role in achieving energy security and sustainable development. This suggests taxation and green support indicators may have influence.

Showing in Fig. 3 risk management during digital transformation is critically important for the economic security of enterprises. Through literature analysis, research has found that studies related to digital transformation and industrial economic security primarily focus on Industry 4.0 and smart manufacturing. Key terms such as digital twins, AI, the Internet of Things, and sustainability have formed a cluster of eight research directions. Against the backdrop of industrial transformation and the development needs of Industry 4.0, scholars such as Kamble, S.S., et al. [24–31] have conducted research on strategic risk management in enterprises through performance evaluation, index analysis, and empirical case studies. Their study highlights that cost, sustainability, and production efficiency should constitute the performance evaluation framework for new smart manufacturing. The implementation of risk control plans and risk management research must be grounded in documented records and rigorous data collection. Methods such as TOPSIS-based MCDM can be employed to evaluate the application of artificial intelligence in Industry 4.0 technologies, some studies have also focused on the safety and economic assessment of the residual lifespan of nuclear power plant equipment, employing Monte Carlo simulation methods and using the Analytic Hierarchy Process (AHP) to calculate economic impact parameters, thereby addressing uncertainties. Additionally, some studies proposes leveraging advanced education to build a modern IoT system and assess the level of industrial manufacturing.

As is illustrated in Fig. 4, marketing and financial management in energy management are crucial for sustainable development. It is essential to assess whether they can

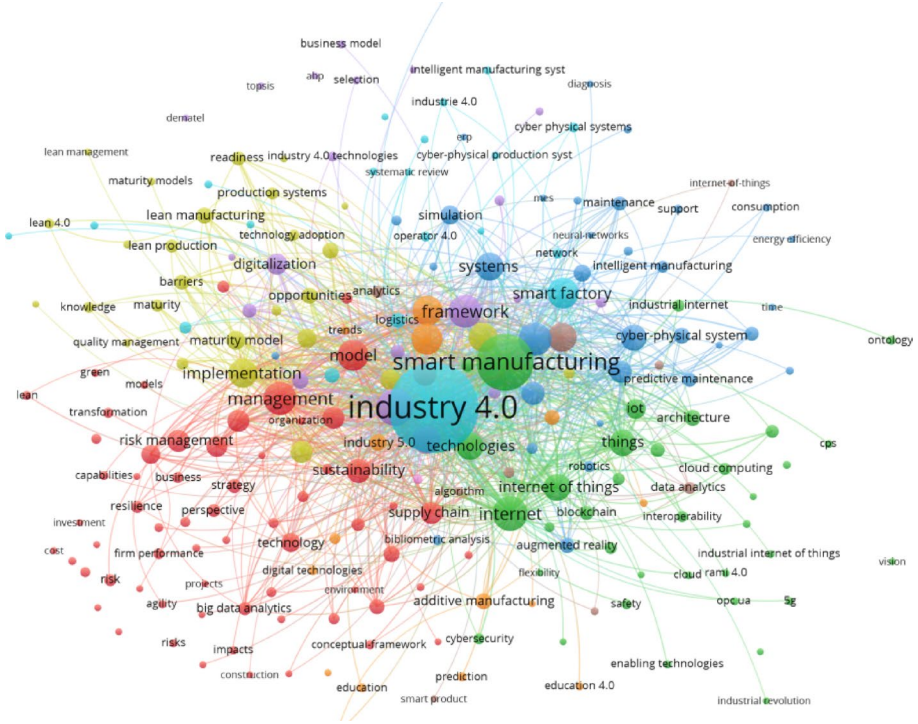


Fig. 3 Analysis of key terms in SCI papers on strategic risk management for digital transformation in the power industry

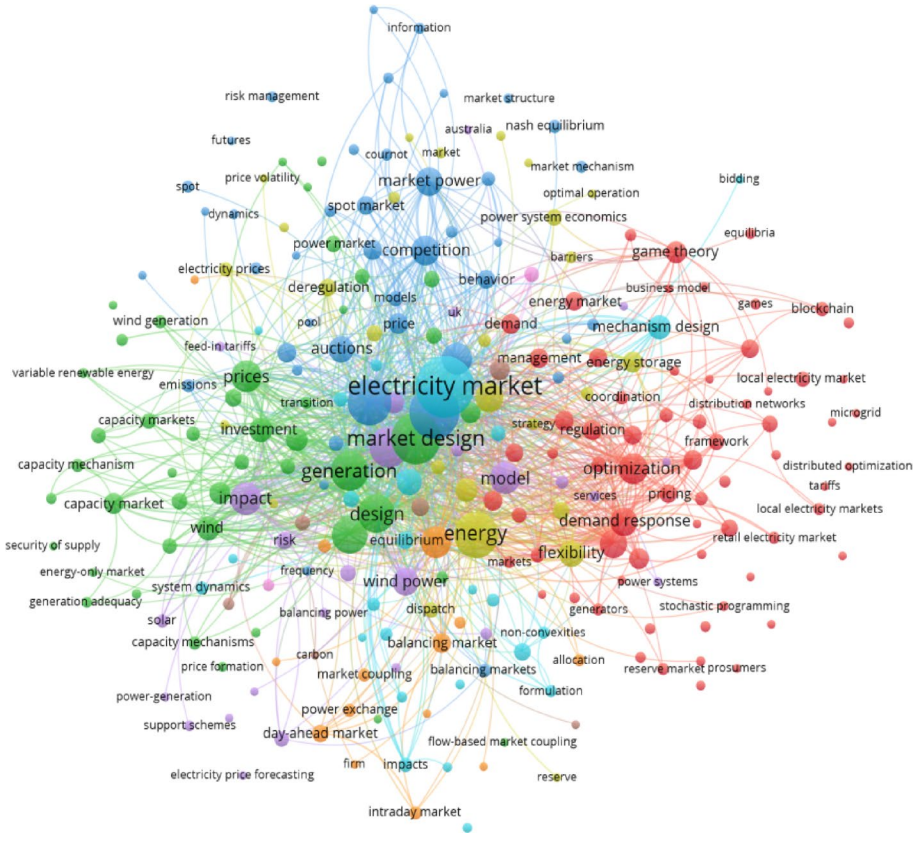


Fig. 4 Keywords network visualization of SCI research papers on energy system marketing and financial management

generate significant benefits for the company's future production and promote its sustainable growth, making it necessary to understand marketing and financial management within corporate management. Through literature analysis, research has found that studies based on the electricity market primarily focus on various issues such as market strategy design, operational demands, and production and transformation optimization. This has formed a cluster of eight research directions, with three main clusters centered around market optimization, market prices, and market competition, while energy, impact, and strategies serve as secondary research focuses.

Steinbach, S.A. et al. [32–38] highly emphasize the importance of focusing on market development policies in their research. For instance, in the context of European hydrogen energy trading, they propose tailoring green energy regulatory schemes around market development policies, infrastructure regulations, and hydrogen and certificate trading. Regarding the transformation of resource structures in the electricity market, scholars have analyzed U.S. power companies and proposed 11 electricity market design schemes for the clean energy transition. These schemes were compared and analyzed based on 10 criteria, including clean energy capacity and market performance, exploring the synergistic effects between long-term hedging requirements and incentives for real-time demand participation. Other scholars have introduced a universal analytical method to dissect these market systems by comparing descriptions of market models, market structures, market timing, and implementation approaches, thereby understanding the interaction mechanisms between future and existing markets. In examining

strategic risk management for industrial enterprises' economic security amid digital transformation, the author developed an integrated digital security management platform framework. This model combines risk analysis, cybersecurity, big data, and AI technologies, while proposing a risk quantification approach based on scenario analysis, expert evaluation, and mathematical modeling. The study highlights the synergistic relationship among strategic management, digital transformation, and economic security, emphasizing that establishing transparent, automated, and responsive risk management platforms is crucial for ensuring corporate continuity and national economic security during wartime or uncertain environments.

From the above research and summary, it can be observed that current studies are relatively comprehensive in terms of both the assessment of mechanical lifespan and research on the economics of electricity markets. Moreover, many advanced research methods have already been proposed. However, it is also evident from the research that while there are studies on the evaluation of sustainable development in power systems, research exploring the relationship between sustainable development and production-marketing capabilities is limited. The theoretical and practical significance of this study can be fully affirmed.

Additionally, based on the aforementioned literature review, it is clear that the selection of the indicator system for this study should take into account both economic factors, such as production capacity and economic indicators, and the sustainable development requirements of Industry 4.0, such as technology-related patents and low-carbon indicators aligned with the carbon peaking and carbon neutrality goals. In terms of methodology, techniques such as Topsis, MCDM, and many others have been applied in related fields, and suitable methods can be selected and adapted based on the characteristics of this study.

3 Methodology and data

3.1 Method selection

The Business Capacity and Sustainable Production Ability of Chinese electric power enterprises involves many factors.

First, a suitable method is needed to carry out calculate the performance. The are methods like AHP, FCE, Entroy-Topsis, one should be chosen for this research.

As for AHP, it is one of the essential for different evaluation. As noted by prior researchers, the weights of relevant indicators rely on expert judgment, which is suitable for subjective occasions. In research of prioritization for renewable energy sources in India, AHP was used, and the result benefits developing nations and guide policymakers in adopting RES [36].

In Research on coordination of nuclear security risk and facility importance evaluation, FCE(Fuzzy Comprehensive Evaluation) plays an essential role, and the analysis results can serve as a reference for allocating nuclear radiation monitoring resources in various facilities in a reasonable manner [37]. FCE also has the problem of strong subjectivity and requires mass data, whose ambiguity leads to the interpretation difficulty. These are not fit for this study.

Comparing with the above two methods, there also a lot of methods which have their advantages and disadvantages, but among this, Entropy-TOPSIS has its unique advantages and adaptability. Entropy-TOPSIS is an objective comparison and evaluation

method that relies entirely on the inherent characteristics of the data to determine weights. This eliminates subjective bias, ensuring results are more objective and reliable. This aligns well with the project's data source, which is derived entirely from objective reports on official websites. By analyzing positive and negative indicators, it evaluates the strengths and weaknesses of alternative solutions. Its wide application in areas such as enterprise competitiveness analysis and product optimization, further underscores its suitability for this study. However, the method has certain limitations, such as computational complexity, sensitivity to extreme values, and constraints in weight determination. The computational complexity can be easily addressed through modern computational tools, making the process highly convenient.

Given the consideration of both sustainable production capacity and production capacity, and based on the previously defined concepts of Business Capacity and Sustainable Production Ability, a dual-system approach is adopted to establish their relationship. Therefore, the coupling coordination degree model will also be applied in the evaluation process.

Finally, in order to quantitatively illustrate the relevant conclusions, the study will also use mathematical models such as conduction model for verification.

3.2 Data selection of the coupling mechanisms

The indicators chosen for the evaluation of Business Capacity and Sustainable Production Ability should be scientific and realiable. The research data of this project are all derived from some relatively objective data, which collecting from the official website of Shanghai Electric Power Co. Ltd., carbon emission list of Chinese listed companies of *Finance and Economics* magazine and other economic data from website https://www.macrodatas.cn.

Business capacity is primarily assessed based on financial health, cost structure, R&D investment, production scale and efficiency, and asset and liability structure. From these dimensions, a total of 16 indicators, including "Investment Income (C1)", "Net Profit (C2)", "Operating Profit Ratio (C3)", and other 13\ indicators were selected for performance evaluation. The Sustainable Production Ability involves 4 parts, Environmental Comprehensive Performance, Technology Innovation & R&D, Financial & Economic Performance, Operational Efficiency & Asset Utilization. There are 17 indicators here. These are shown below in Table 1.

The above data are collected from official annual ESG report or yearly economic report, years from 2017–2023 was chosen to evaluate for the reason about data in recent years is relatively complete, and the span is also relatively large, which can be used as a reference cycle. In fact, China's economic plan is a five-year cycle, but the impact of the epidemic of Covid-19 is not clearly, so, two more years were chosen as a reference. Specific data for the two systems are shown below appendix.

Regarding the aforementioned indicators, the following supplementary notes are provided. Although the Entropy-TOPSIS method was selected, primarily to account for data information entropy, uniformity in units was maintained during data entry to prevent excessive deviation in research results. For instance, monetary indicators such as operating revenue and net profit are denominated in billions of RMB, while ratio-based data are expressed as decimals rather than percentages.

Table 1 Table of all indicators

Index layer for business capacity	Index layer for sustainable production ability
Investment income C1	Huazheng ESG score annual average F1
Net profit C2	Total investment in environment protection F2
Operating profit ratio C3	Total carbon emissions F3
Operating revenue C4	Proportion of clean energy installed capacity F4
Return on equity C5	Number of patents granted and under review F5
Operating cost C6	Research and development expenses F6
Research and development expenses C7	Total number of R&D personnel F7
Number of employees C8	R&D investment amount F8
Installed capacity C9	Investment income F9
Electricity generation (Domestic) C10	Net profit F10
Debt-to-asset ratio C11	Return on assets (TTM method) F11
Current asset ratio C12	Investment yield F12
Fixed asset ratio C13	Total assets F13
Total assets C14	Digitalization level F14
Total liabilities C15	Total factor productivity (LP method) F15
Total debt ratio C16	Installed capacity F16
	Electricity generation F17

The dataset contains few missing values, but these can't be overlooked. To minimize their impact on the results, a reasonable approach is primarily adopted for handling them. This involve combining linear trend extrapolation and growth rate methods, supplemented by trend-based judgment. For example, when processing the F2 data of the comparative company Huadian Power International, the value for 2017 was missing. Three data points were inferred using this methodology, C7 for 2017 of Shanghai Electric Power Co., Ltd., F2 for 2023 of Shanghai Electric Power Co., Ltd., F6 for 2017 of Huadian Power International Corporation Limited(short for HPI). The calculation process is shown in the appendix.

3.3 Research practice process

The two systems will use the Entropy-TOPSIS method to judge the weight of each index. Details are as follows:

1) Entropy-TOPSIS Evaluation: Creating a decision matrix X where rows represent alternatives and columns represent criteria. Let there be m alternatives and n criteria.

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix} \tag{1}$$

Here, x_{ij} is the performance value of the i-th alternative with respect to the j-th criterion. Then Normalizing the decision matrix to eliminate the influence of different units. For positive index (higher is better) and negative index (lower is better), use:

$$r_{ij} = \begin{cases} \frac{x_{ij} - \min\{x_j\}}{\max\{x_j\} - \min\{x_j\}} \ (x_{ij}\ is\ a\ positive\ index) \\ \frac{\max\{x_j\} - x_{ij}}{\max\{x_j\} - \min\{x_j\}} \ (x_{ij}\ is\ a\ negative\ index) \end{cases} \tag{2}$$

Subsequently, a matrix R was obtained after standardization, the information entropy of each index is E_j:

$$E_j = -\frac{1}{\mathrm{In}\ (m)} \sum\nolimits_{i=1}^{m} p_{ij} \ln (p_{ij}) \tag{3}$$

where p_{ij}:

$$p_{ij} = \frac{r_{ij}}{\sum\nolimits_{i=1}^{m} r_{ij}} \tag{4}$$

In the formula, E_j is the information entropy of index j, p_{ij} is the proportion of j-th index of i-th evaluation object in the overall index, determining the weight of each index:

$$w_i = \frac{1 - E_j}{\sum\nolimits_{j=1}^{n} (1 - E_j)} \tag{5}$$

The weights satisfy $\sum\nolimits_{j=1}^{n} w_j$=1.

Multiply the normalized matrix R by the entropy weights w_j, get matrix V.

Compute the Euclidean distance of each alternative from the ideal solution A^+ and the negative-ideal solution A^-:

Distance to ideal solution and distance D_i^+ to negative-ideal solution D_i^-, Compute the relative closeness C_i of each alternative to the ideal solution:

$$C_i = \frac{D_i^-}{D_i^- + D_i^+} \tag{6}$$

Rank the alternatives based on the relative closeness C_i. The higher the C_i, the better the alternative.

2) Coupling Coordination Analysis: Coupling analysis models are applicable when there needs to be exchange or mutual influence of information, energy, and materials between two systems, such as the interaction between power systems and information systems, or the coupling between power grids and transportation networks [39]. The structure and behavior of these systems are relatively complex, making it difficult to accurately describe and analyze them using a single model or method. Therefore, coupling analysis models are required to comprehensively consider the multifaceted characteristics of the system. Coupling analysis models can more fully reflect how interactions between systems affect overall performance [40]. The coupling relationships between systems exhibit different characteristics at different time scales, for example, the coupling relationship between power systems and transportation systems under extreme disaster conditions can vary before, during, and after the disaster. In such cases, coupling analysis across multiple time scales is necessary. This study In fact, it is related to the production problem of power companies, and lies in the relationship between production profit and sustainable development, and it examines the relationship between the two systems, so the coupling coordination degree is suitable.

Coupling refers to the process in which the interaction between two or more elements affects each other. It has a certain universality, it is often used by the economic circle to judge the order of the development of variables. Thus, it is possible to use this method



Fig. 5 Classification of coupling degree between two system

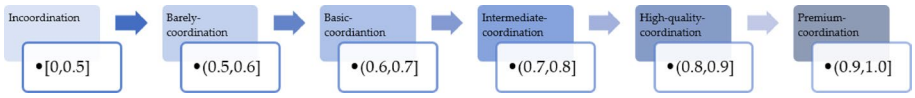


Fig. 6 Classification of coupling coordination degree between two systems

to measure coordination relationship between Business Capacity and sustainable to make decisions for future development. First, the index system is divided according to the two systems of Business Capacity and Sustainable Production Ability. An electric power company is a huge and complex system, which has a lot of departments and machines, and its production ability and development situation is important essential factors of survival. The coupled coordination model can reflect the degree of coordination between the systems and reflect the stage of the level of coordination development [41]. Since this study has only two systems, so the Business Capacity is W_1, Sustainable Production Ability is W_2, the specific formula is as follows:

$$C = 2\sqrt{f(W_1)f(W_2)/(f(W_1) + f(W_2)} \tag{7}$$

$$T = \alpha f(W_1) + \beta f(W_2) \tag{8}$$

$$D = \sqrt{CT} \tag{9}$$

In the formula, C is stand for the coupling degree between the the Business Capacity and Sustainable Production Ability, $f(W_1)$ and $f(W_2)$ are the comprehensive evaluation index of the two respectively. The coupling degree between the two systems can be judged by the size of C. And T is a comprehensive coordination degree of the two systems, α is the contribution share to resilient urban systems, β is contribution share to the smart city system, D is the coupling coordination degree of the two systems. In fact, the Business Capacity and Sustainable Production Ability are both important, thus, it can be said that they are of equal importance. So, in this research, $\alpha = \beta = 0.5$. As for the level of the coupling degree, different experts hold on different opinions. Liu, Q [42]. classify 6 levels for the coupling degree and coupling coordination degree of agricultural products and resource environment. In order to avoid complicated work in research in this project, six levels (Figs. 5 and 6) are selected for reference.

Therefore, this paper will first use the Entropy-TOPSIS method to calculate the weights of indicators, and then use the coupled coordination model for analysis. After that, in order to identify the main obstacle subsystems and the main factors of obstacles, an analysis of the main obstacle factors in the system will be conducted by using the Obstacle Degree.

3) Obstacle Degree Analysis: The Obstacle Degree is a measure used to identify the extent to which each criterion acts as an obstacle or hindrance to the overall performance of an alternative. It is often used in decision-making processes to diagnose and improve

system performance. Below is the formula and calculation process for the Obstacle Degree, for each criterion *j*, compute the deviation degree $d_{ij}$for each alternative *i*. The deviation degree represents how far the performance value of a criterion is from the ideal value.

$$d_{ij} = \left| x_{ij} - x_j^* \right| \tag{10}$$

where:

x_{ij}: The performance value of the *i*-th alternative for the *j*-th criterion.

x_j^*: The ideal value for the *j*-th criterion (e.g., the maximum value for benefit criteria or the minimum value for cost criteria).

Normalize the deviation degree to ensure comparability across criteria:

$$d_{ij}^{'} = \frac{d_{ij}}{\max\left(d_{ij}\right)} \tag{11}$$

where max (d_{ij}) is the maximum deviation degree for the *j*-th criterion across all alternatives.

The obstacle degree O_{ij} for the *i*-th alternative and the *j*-th criterion is calculated as:

$$O_{ij} = \frac{w_j d_{ij}^{'}}{\sum_{j=1}^{n} w_j d_{ij}^{'}} \tag{12}$$

where:

$d_{ij}^{'}$: The normalized deviation degree.

n: The total number of criteria.

To evaluate the overall obstacle degree for each criterion across all alternatives, aggregate the obstacle degrees:

$$O_j = \sum_{i=1}^{m} O_{ij} \tag{13}$$

where:

O_j: The overall obstacle degree for the *j*-th criterion.

m: The total number of alternatives.

The obstacle levels are divided to 6 levels, as is shown in Table 2.

4) Hypothesis-Driven Mathematical Modeling: after analyzing the relevant results, considering the effects of transformation in the process of sustainable development and the impact of the epidemic during this period, in order to make the results more scientific, the fixed effects model is used to verify the impact of green transformation and the impact during the epidemic period, so as to better quantify the effects of industrial transformation. The formula is as follows:

Table 2 Main factor obstacle degree of business capacity and sustainable production ability

Proportion	Obstacle levels
[0, 10%)	Obstacle-free (0) or Low-obstacle level
[10%, 40%)	Mild-obstacle level
[40%, 60%)	Moderate-obstacle level
[60%, 90%)	High-obstacle level
[90%, 100%]	Extreme-obstacle level

$$\begin{aligned}\Delta Perf_{it} =& \beta_0 + \beta_1 Covit_t + \beta_2 Sanction_t \\ &+ \beta_3 SR_i + \gamma\left(SR_i \times Covit_t\right) \\ &+ \delta\left(SR_i \times Sanction_t\right) + \theta Controls_{it} \\ &+ \mu_i + \delta_t + \varepsilon_{it}\end{aligned} \tag{14}$$

Variable definition and assignment therein are there in Table 3:

In this process, the relevant data is chosen to input into Deepseek, and ChatGLM for computation. This approach not only leverages the computational power of large models to enhance processing speed but also improves the accuracy of the results. Meanwhile, by conducting a comprehensive comparison of the outcomes, the differences is analyzed and underlying reasons among the results generated by these AI systems, ultimately leading to a definitive conclusion.

To analyze the applicability of the modeling methodology, the results of each phase—after completing the assessment for Shanghai Electric Power Co., Ltd.—are compared with HPI as the reference group. This comparative analysis facilitates a discussion on the replicability of the research and deepens the examination of practical implications in the energy sector.

4 Results

4.1 Results of performance for two systems

By using the Entropy-TOPSIS to judge, after evaluation, after calculating, the AI has given the result, then checking computations according to the previous evaluation methods and processes, it shows the same result. So, the weight of each index is as follow Tables (Tables 4 and 5).

The Entropy-TOPSIS results reveal fluctuating performance in both Business Capacity and Sustainable Production Ability over the study period. As shown in the bar chart(Figs. 6 and 7), 2023 recorded the highest scores for both systems.

For Business Capacity, 2021 was the weakest year, with a score of only 0.3369. In contrast, 2023 achieved the highest value of 0.7080—more than double that of 2021. Throughout the period, Business Capacity initially experienced a slight increase, followed by a gradual decline during the early and peak phases of the pandemic from 2019–2021. It then rebounded sharply, reaching its peak by the end of 2023.

Table 3 Variable definition and assignment table

Variable notation	Definition, and assignment description
$\Delta Perf_{it}$	Change in firm performance: the growth rate of total profit (C2) for firm i in year t relative to year t–1
$Covit_t$	Pandemic dummy variable: equals 1 if year t is 2020 or 2021, and 0 otherwise
$Sanction_t$	Intensity of economic sanctions: the inverse (or negative) of the growth rate of total import and export in t years is used as an agent variable. The lower the growth rate, the greater the intensity of sanctions
SR_i	Smart resilience index: Core variable. It is measured by the comprehensive index of entropy weighting of digital level (F14) and R&D investment intensity (F6/Total Assets). The average value of i enterprise from 2017 to 2019 represents its pre-existing smart transformation capability.
Interaction Terms	Interactive item: $SR_i \times Covit_t$ and $SR_i \times Sanction_t$ and the core parameters to be estimated γ and δ. This is called the moderating effect of wisdom resilience.
$Controls_{it}$	Control variables: enterprise size (Total Assets, C14), asset-liability ratio (Debt-to-asset Ratio, C11)
μ_i, d_t	Enterprise individual effect and year fixed effect

Table 4 Weight table of business capacity performance evaluation of Shanghai electric power Co., ltd

Index layer C	The weight of C to A	+(positive)/ -(negative)
Investment income C1	9.11%	+
Net profit C2	3.22%	+
Operating profit ratio C3	3.12%	+
Operating revenue C4	6.92%	+
Return on equity C5	3.34%	+
Operating cost C6	5.69%	−
Research and development expenses C7	14.92%	+
Number of employees C8	5.13%	+
Installed capacity C9	5.92%	+
Electricity generation (Domestic) C10	6.07%	+
Debt-to-asset ratio C11	4.42%	−
Current asset ratio C12	5.02%	+
Fixed asset ratio C13	5.65%	+
Total assets C14	6.00%	+
Total liabilities C15	9.65%	−
Total debt ratio C16	5.81%	−

Table 5 Weight table of sustainable production ability performance evaluation of Shanghai electric power Co., ltd

Index layer F	The weight of F to D	+(positive)/ -(negative)
Huazheng ESG score annual average F1	3.77%	+
Total investment in environment protection F2	4.94%	+
Total carbon emissions F3	5.66%	−
Proportion of clean energy installed capacity F4	3.56%	+
Number of patents granted and under review F5	11.21%	+
Research and development expenses F6	5.62%	+
Total number of R&D personnel F7	8.11%	+
R&D investment amount F8	12.89%	+
Investment income F9	9.00%	+
Net profit F10	3.18%	+
Return on assets (TTM method) F11	3.42%	+
Investment yield F12	9.73%	+
Total assets F13	4.96%	+
Digitalization level F14	2.99%	+
Total factor productivity (LP method) F15	5.89%	+
Installed capacity F16	5.07%	+
Electricity generation F17	5.85%	+

As for Sustainable Production Ability, the overall trend was similar, though the lowest score occurred in 2017 at 0.3071, while the highest was also in 2023 at 0.5767—nearly 1.9 times that of 2017. The trajectory of Sustainable Production Ability was relatively smoother: a substantial rise from 2017–2018 established a solid foundation, followed by only a minor dip in 2019 and 2020. A steady recovery began in 2021, culminating in the 2023 score of 0.5767.

As a control group, relevant data on HPI were also collected, and through the same calculation, the calculation results of performance evaluation in Tables 10 and 11 were finally obtained, the final scores of the comprehensive evaluation yielded the results shown in Figs. 5 and 6. Subsequently, a coupling coordination analysis was conducted on



Fig. 7 Performance result of business capacity on Shanghai electric power Co., Ltd

Table 6 Weight table of business capacity performance evaluation of HPI

Index layer C	The weight of C to A	+ (positive)/ – (negative)
Investment income C1	11.47%	+
Net profit C2	3.22%	+
Operating profit ratio C3	3.24%	+
Operating revenue C4	6.18%	+
Return on equity C5	3.22%	+
Operating cost C6	7.15%	–
Research and development expenses C7	8.17%	+
Number of employees C8	7.95%	+
Installed capacity C9	12.03%	+
Electricity generation (Domestic) C10	4.46%	+
Debt-to-asset ratio C11	3.89%	–
Current asset ratio C12	4.79%	+
Fixed asset ratio C13	6.97%	+
Total assets C14	8.76%	+
Total liabilities C15	4.49%	–
Total debt ratio C16	4.04%	–

the results, leading to Fig. 7. Finally, an obstacle degree analysis was performed, resulting in Tables 6 and 7.

Figure 9 (Business Capacity) rose monotonically from 0.3314 in 2017 to 0.6708 in 2023, averaging 12.46% annual growth and a 102.40% total increase, with the steepest gains recorded after 2020.

Figure 10 (Sustainable Production Ability) hit a trough of 0.1747 in 2017, peaked at 0.7236 in 2023, and followed a V-shaped path: a steady decline through 2020 (bottom 0.2664) and an accelerated rebound thereafter, yielding a 314% cumulative gain.

4.2 Coupling coordination of two systems

Then the coupling degree and coupling coordination degree can be calculated from the above data. According to the calculation of coupling coordination model, the coupling degree and coupling coordination degree curve of Shanghai Electric Power Co., Ltd.'s Business Capacity and Sustainable Production Ability have been illustrated. The result

Table 7 Weight table of sustainable production ability performance evaluation of HPI

Index layer F	The weight of F to D	+ (positive)/ – (negative)
Huazheng ESG score annual average F1	3.25%	+
Total investment in environment protection F2	12.32%	+
Total carbon emissions F3	5.74%	–
Proportion of clean energy installed capacity F4	2.75%	+
Number of patents granted and under review F5	4.22%	+
Research and development expenses F6	7.64%	+
Total number of R&D personnel F7	5.70%	+
R&D investment amount F8	4.94%	+
Investment income F9	10.73%	+
Net profit F10	3.01%	+
Return on assets (TTM method) F11	2.98%	+
Investment yield F12	6.17%	+
Total assets F13	4.94%	+
Digitalization level F14	4.70%	+
Total factor productivity (LP method) F15	5.50%	+
Installed capacity F16	11.26%	+
Electricity generation F17	4.17%	+



Fig. 8 Performance result of sustainable production ability on Shanghai electric power Co., Ltd



Fig. 9 Performance result of Business Capacity on HPI



Fig. 10 Performance result of Sustainable Production Ability on HPI

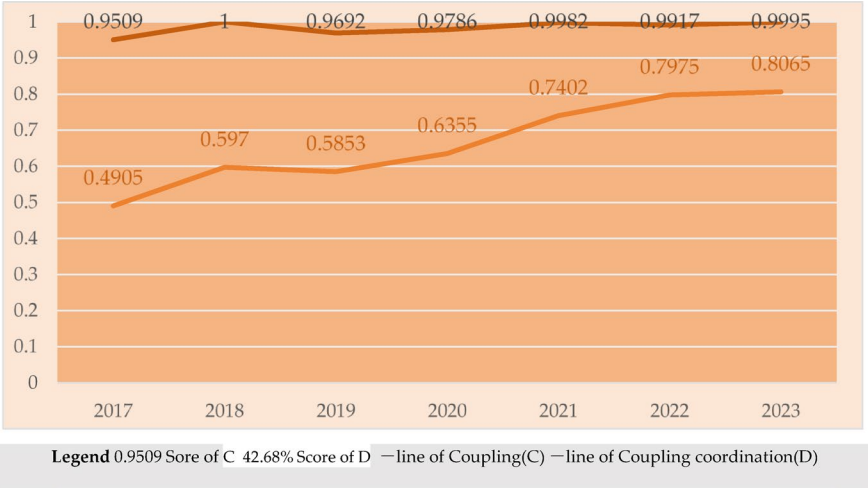


Fig. 11 Coupling coordination degree of each year for the two systems

can be seen at Fig. 11. It shows that from 2017–2023, the coupling degree between Business Capacity and Sustainable Production Ability of Shanghai Electric Power Co., Ltd. has always been high, with the coupling degree basically above 0.99, which belongs to the high-level. The annual change in the coupling degree between the two systems is small, all less than 0.007. But for coupling coordination degree, it is in different situation. Overall, there is a positive interactive and influential relationship between the two systems. Looking at the coupling coordination degree curve, the first two years showed an obvious increase and climb to 0.71052, then a smoothly decrease happened from 2018–2021, and the score dropped to 0.61893 reaching the bottom. After that, it is getting better. The score increase to the peak at 2023, whose number is 0.79935. It can be summarized that the coupling degree of the two systems is quite good, the coupling coordination degree slightly fluctuate, but finally reach the peak.

As for the main obstacle indicators of two systems, guided by the Pareto Principle, this study focuses on the top three barrier indicators, identified as the key factors comprising the top 20%. However, the analysis is appropriately broadened in scope to ensure its scientific rigor. Business Capacity (Table 8), the analysis can be divided into 2 stages the first 5 years and the recent 2 years for the reason that early work and recent difficulties

Table 8 A list of the main obstacle indicators in business capacity

Year	Rank 1	Rank 2	Rank 3
2017	C8/22.18%	C4/10.49%	C1/10.03%
2018	C8/28.98%	C12/11.79%	C4/11.31%
2019	C8/25.48%	C1/14.19%	C4/9.76%
2020	C8/23.74%	C1/12.90%	C15/8.84%
2021	C8/16.48%	C15/14.55%	C1/13.74%
2022	C15/19.32%	C1/16.41%	C8/14.34%
2023	C15/32.51%	C11/20.53%	C1/19.62%

Table 9 A list of the main obstacle indicators in sustainable production ability

Year	Rank 1	Rank 2	Rank 3
2017	F8/18.61%	F7/11.71%	F9/9.43%
2018	F8/26.28%	F7/15.25%	F6/11.12%
2019	F8/21.89%	F12/15.92%	F9/15.07%
2020	F8/18.93%	F5/17.16%	F12/14.22%
2021	F5/18.60%	F12/16.31%	F9/15.09%
2022	F5/22.40%	F12/17.70%	F9/15.60%
2023	F8/22.88%	F5/20.54%	F12/15.58%

can be analyzed. In order to find the main obstacle indicators, all indicators are ranked, and the top 3 was recorded in Tables 10 and 11. Shown in the below table, C8 has always been the top indicator of the first five years and the third of 2022, so great attention should be paid in C8. Then, C1 is an indicator that can not ignore in the research, because it ranked 3 in 2017 and 2021, 2 in 2019 and 2020, and in recent 2 year, it still rank 2 in 2022 and 3 in 2023. Besides, C15 should also be watched, in 2020 and 2021, it ranked 3 and 2 respectively. And in recent 2 years, C15 stand at the top. C11 is emerging high obstacle indicator. C4 ranked 3 in 2018 and 2019.

The main obstacle indicators of Sustainable Production Ability is also obvious (Table 9). The highest attention should be paid at F8. From 2017 to 2020, F8 is the top obstacle, and also top in 2023. Another indicator that should be concerned is F5, which has a very high rank in recent 4 years, for the rank of which is top in 2021 and 2022, and rank 2 in 2020 and 2023. Then is F12, it can be seen that it shows up in 2019 at the rank of 2, and for the next 4 years, its rank has also changed between 2 and 3 places. The next indicator that should be noticed is F9, in 2017, 2019, 2021 and 2022, the ranking has been 3. Except for the above indicators, F7 ranked 2 in 2017 and 2018, F6 ranked 3 in 2018.

As for HPI, applying the coupling-coordination model to the same dataset yields the curves displayed in Fig. 7. Over the 2017–2023 window, the coupling degree between Business Capacity and Sustain.

Sustainable Production Ability remains persistently high, hovering between 0.9509 and 0.9995 and staying within an extremely narrow annual bandwidth of 0.007, indicating a stable, high-level resonance between the two subsystems. In contrast, the coupling coordination degree follows a markedly different trajectory: it climbs steeply from 0.4905 in 2017 to an early plateau of 0.597 in 2018, then falls to 0.5853 in 2019. Thereafter, the index reverses course and ascends continuously, culminating at an all-time high of 0.8065 in 2023. Thus, while the systems maintain near-optimal coupling throughout the observation period, their coordination experiences a mild but discernible fluctuation before ultimately reaching peak harmony.

Table 10 A list of the main obstacle indicators in business capacity

Year	Rank 1	Rank 2	Rank 3
2017	C9/18.00%	C1/16.10%	C14/13.09%
2018	C9/18.40%	C1/16.87%	C14/12.73%
2019	C9/20.56%	C1/19.85%	C14/14.33%
2020	C1/22.83%	C8/14.49%	C14/13.71%
2021	C6/14.76%	C8/14.49%	C13/14.39%
2022	C8/18.02%	C13/15.63%	C6/14.20%
2023	C8/23.98%	C6/20.53%	C1/18.10%

Table 11 A list of the main obstacle indicators in sustainable production ability

Year	Rank 1	Rank 2	Rank 3
2017	F1/14.51%	F16/13.64%	F9/12.20%
2018	F16/17.29%	F9/15.85%	F6/11.42%
2019	F1/15.54%	F16/14.67%	F9/14.16%
2020	F1/18.33%	F9/15.96%	F16/15.82%
2021	F1/19.36%	F3/10.82%	F8/10.15%
2022	F9/14.14%	F12/11.73%	F13/11.07%
2023	F1/32.97%	F9/15.06%	F7/14.69%

Table 10 pinpoints the dominant bottlenecks within Business Capacity. C9 commanded the greatest share at the outset, consecutively topping the list in 2017–2019 with an escalating weight that peaked at 20.56%. C1 then overtook C9, becoming the primary obstacle in 2020 (22.83%) while still retaining a top-three position in every subsequent year and ranking third in 2023 (18.10%). After 2020, attention must shift to C8 and C6: C8 moved from second place in 2020–2021 to the foremost barrier in 2023, accounting for almost a quarter of all obstacles (23.98%), whereas C6 emerged second in 2021, maintained third in 2022, and leapt to rank two in 2023 (20.53%). C14 consistently lingered among the top three between 2017 and 2020, gradually easing its share from 13% to 14% before dropping out of the leaderboard. Finally, C13 surfaced in 2021–2022, securing third and second places respectively, signalling a transient but notable constraint. Collectively, these indicators reveal an evolving hurdle profile that has shifted from C9/C1 dominance pre-2020 to a C8/C6-centric challenge in the most recent period.

The key barriers impeding Sustainable Production Ability are clearly identifiable from the table. F1 demands the greatest attention: it occupied the top rank in 2017, 2019–2021 and, most strikingly, accounted for almost one-third of total obstacles (32.97%) in 2023. F9 is another persistent hindrance, appearing among the top-three factors every single year and holding the No. 2 position in 2018 and 2023 while contributing roughly 15% in those years. F16 also figures prominently, ranking second in 2017–2019 and third in 2020, with obstacle shares consistently above 13%. After these three, F12 and F13 emerged as new focus areas after 2020: F12 ranked second in 2022 (11.73%) and F13 third in the same year (11.07%), signalling a recent shift in the constraint structure. Other sporadic but noteworthy impediments include F6 (rank 3 in 2018), F3 (rank 2 in 2021), F8 (rank 3 in 2021) and F7 (rank 3 in 2023), all of which briefly entered the top-three list and should therefore remain under surveillance (Table 12).

4.3 Comparison of Shanghai electric power Co., Ltd. And HPI results

The numerical results show that both companies maintained an overall upward trajectory in their performance evaluation, and the coupling-coordination degree of each

Table 12 Expected regression results table

variable	Coefficient expectations	t	Economic meaning
$Covit_t$	$\beta_1 = -0.35^{***}$	− 3.7	The epidemic has reduced the growth rate of corporate profits by 35% points, with a significant impact.
$Sanction_t$	$\beta_2 = -0.68^{**}$	− 3.1	**For every 1 unit increase in the intensity of economic sanctions (that is, a deviation from the expected value of 100%)**, the growth rate of corporate profits drops by 68% points. **The impact is very severe.**
SR_i	$\beta_3 = -0.10^{*}$	1.4	Wisdom resilience itself has a positive impact on enterprise performance, but it is not statistically significant.
$SR_i \times Covit_t$	$\gamma = 0.22^{***}$	3	**Core finding 1**: Smart resilience significantly buffers the impact of the epidemic. Enterprises with high resilience reduce the negative impact of the epidemic by 22% points.
$SR_i \times Sanction_t$	$\delta = 0.40^{**}$	3.3	**Core finding 2**: Smart resilience significantly buffers the impact of economic sanctions. Enterprises with high resilience reduce the negative impact of sanctions by 40% points. The buffering effect is very prominent.
$Controls_{it}$	…	…	It has been controlled
R^2	0.78		The model is interpretable

system ultimately climbed to a higher level. The HPI's Business Capacity follows a steady, almost linear ascent, whereas Shanghai Electric Power Co., Ltd. experienced a volatile, step-wise rise; likewise, the nadir of its Sustainable Production Ability occurred in 2019, diverging from the valley year observed for Shanghai Electric Power. These contrasts suggest that the two companies should adopt differentiated road-maps for future growth. Turning to coupling-coordination, HPI recorded only a minor dip in 2019, whereas Shanghai Electric Power reached a local peak in 2018, declined for several years, and did not resume its climb until after 2021. It is evident that the overall trends of both exhibit strong similarity, yet discrepancies exist in the specifics.

The leading obstacle factors overlap to some extent—C1, C8 and F9 appear on both lists—indicating these three indicators are likely to be universal obstacle indicators. In addition, F12 consistently ranks within the top five in the HPI. Although this position is lower compared to Shanghai Electric Power Co., Ltd., which is perennially in the top 2 or 3, F12 remains a universal obstacle indicators. However, a marked divergence exists in the distribution of their barrier factors. Some are the same. Nevertheless, the differences are conspicuous. Shanghai Electric Power Co., Ltd. faces obstacle indicators regarding Business Capacity, with C4 and C15 being the most common. In contrast, the prevailing factors for HPI are C6, C9, and C14. As for Sustainable Production Ability, F5, F7, and F8 are factors that Shanghai Electric Power Co., Ltd. should prioritize individually, whereas F1 and F12 is unique to HPI.

5 Discussion

China vigorously advocates new energy and artificial intelligence drive, and in order to achieve Industry 4.0, complete the industrial transformation, electric power industry also spare no effort to promote the relevant work. Shanghai Electric Power Co., Ltd. is an advanced energy company that unites conventional energy with clean energy, while also focusing on the R&D and application of smart energy technologies. By embracing the principles of "innovation, coordination, green, open, sharing" the company drives its high-quality, sustainable growth. With its goals and values guiding its trajectory,

it fosters a positive corporate culture that contributes significantly to its long-term progression.

In general, as the result shown above, combined with data collected, it is easy to conclude that Business Capacity and Sustainable Production Ability of Shanghai Electric Power Co., Ltd. waves a lot and suffered by the epidemic of COVID-19 in 2020 and 2021, but finally achieve a great progress in 2023. Through years of hard work, Shanghai Electric Power Co., Ltd. realize the balance of Business Capacity and Sustainable Production Ability at level of Intermediate-coordination at a score of 0.7994, basically close to high-quality-coordination. That is to say in the field of intelligent management, intelligent manufacturing, low-carbon production, digital transformation, R&D and continuous increase in production has achieved great progress.

So how is the relationship in Business Capacity and Sustainable Production Ability of Shanghai Electric Power Co., Ltd.? From the rank of obstacle indicators shown above in 2017–2023, combining with the coupling coordination performance result, problems can be easily figure out.

5.1 Solid foundation: pre-pandemic strategic layout and initial success

In 2017, Shanghai Electric Power Co., Ltd. carried out the Industry 4.0 solution plan and fully promoted green manufacture, vigorously promoted digital transformation, enhanced the talent cultivation and cooperation, participated in industry policy designation and strengthen industry influence. This year, by taking the above actions, It has increased the investment in industry 4.0 technology, comprehensively improved the equipment monitoring, automation control and data analysis, and introduced the advanced industrial information system to improve the production efficiency and management level of enterprises. In this year, big data and IOT are applied to support the comprehensive monitoring system of the supply chain. Basic on AI decision support system, better problem analyzing, operational processes optimizing, and data-driven decisions making have done. In terms of cooperation, the development of new energy projects and the cooperation in intelligent manufacturing have been carried out. Shanghai Electric Power Co., Ltd. continued to promote a number of whole county development projects, laying a solid foundation for low-carbon sustainable intelligent manufacturing. In addition, Shanghai Electric Power Co., Ltd. actively participated in the formulation of relevant industry standards and technical specifications, providing strong support for the development of the field of intelligent manufacturing.

Analogously, HPI's 2017 economic report reveals that the national coal-capacity cut immediately drove up coal prices and eroded the economics of conventional thermal power. In response, the company instituted quota management, benchmark-based control and rigorous process oversight to enforce lean cost management and curb expenditures. At the same time, HPI accelerated its clean-energy pivot: by year-end its renewable portfolio—comprising hydro, wind, solar and biomass—had reached 5,753.6 MW of installed capacity across 29 controlled generating units, all supported by big-data analytics and advanced energy-technology platforms. Beyond hardware, the firm embedded an eco-centric culture, stocking rare fish in river-basin rehabilitation programmes, planting endangered seedlings under its "Low-Carbon Love" afforestation campaign and engaging staff in "Small Family, Big HPI" team-building initiatives that link environmental

stewardship with corporate identity. These simultaneous investments in talent and ecology have laid a resilient foundation for HPI's future growth.

It is really a good start in 2017, what the company has done cannot neglect of this year. focusing on technological innovation, digital transformation and cooperation expansion, Shanghai Electric Power Co., Ltd. has comprehensively promoted industrial transformation and intelligent manufacturing. This series of measures reflects the strategic layout of Shanghai Electric Power in the era of industry 4.0, showing its forward-looking and leadership in the field of intelligent manufacturing. Although Performance result of both systems in Shanghai Electric Power Co., Ltd. is very low, but it really tried its best to realize the transformation and laid a good foundation. The level of coupling coordination is barely-coordination that tells a situation that the beginning of the transformation has a lot of difficulties, it is the same as all other transformation, but how the outcome depends on what the company does next.

From the main obstacle indicators given above by the tables, investment income (C1), operating revenue (C4), number of employees(C8) are the main obstacle indicators of Business Capacity system in 2017, while R&D investment amount(F8), total number of R&D personnel (F7), investment income (F9) can not be ignored in Sustainable Production Ability system. In 2017, this shows that the company has achieved a relatively stable financial performance in this year, but from the perspective of profit margin, the company's profitability needs to be further improved, and there may be room for cost control and efficiency optimization. The company's scientific research investment in 2017 reflects the importance of Shanghai Electric Power Co., Ltd. in technological innovation and R&D projects. Its active participation in environmental protection and social responsibility projects is also closely related to scientific research investment, such as the development of clean energy technologies or the optimization of energy efficiency to support the city's modernization process, meet market competition needs and support sustainable development goals. However, as the first year of this research, it is predictable that relatively little investment income, profit and research investment become the main obstacles. At the same time, as the initial stage, the reserve of researchers is obviously insufficient, which needs to be adjusted slowly, and the research investment must be gradually enhanced to change the situation.

As a control case, HPI's top obstacles—Investment income (C1), Installed Capacity (C9, F16) and Total Assets (C14, F9)—mirror the systemic shocks that national-level policies inflict on large utilities. The surge in coal prices and the freeze on conventional infrastructure, for a large enterprise like HPI, which is mainly based on traditional thermal power generation, inevitably compress Total Assets. Yet the data show that HPI has already built a credible transition platform: 29 subsidiaries now host clean-energy generators, tangible proof of a forward-looking strategic layout.

5.2 Pandemic impacts and structural challenges

Then the period of 2018–2021, Shanghai Electric Power Co. Ltd. continued to optimize the energy structure from 2018–2021, and the proportion of clean energy continued to increase, rising from 43.48% in 2018 to 55.24% in 2021. The company's assets and installed capacity of holding shares have steadily increased, reaching 156.709 billion yuan and 19.0531 million kw respectively in 2021. The company's operating revenue and power generation maintained growth, reaching 30.631 billion yuan and 59.742 billion

KWH respectively in 2021. The profitability of the company improved, and the total profit increased by 39% in 2021 year on year. The company's credit rating was upgraded, and AAA rated in 2021, with a stable outlook.

During the period, Shanghai Electric Power Co., Ltd. has made significant strides in optimizing its energy structure, improving profitability, protecting shareholder interests, strengthening risk management, deepening the reform of state-owned enterprises, and strengthening its application of technology. These efforts have contributed to the company's Sustainable Production Ability. In 2021, there was a recovery, but Business Capacity continued to decline. This can be attributed to two main factors: the onset of economic sanctions, the severe impact of the pandemic, and the growing pains of industrial transformation. Since economic sanctions began in 2018, a series of trade wars disrupted economic production. Additionally, the core of industrial intelligence and smart production also began to emerge in 2018, leading to a decline in Business Capacity and Sustainable Ability between 2018 and 2019. Subsequently, the pandemic broke out, further limiting production capacity, while industrial transformation continued to advance. The traditional energy-focused power industry began transitioning toward green and intelligent energy development, which sustained the decline in Business Capacity. The pandemic's impact persisted into 2021, and as smart energy initiatives were deeply integrated into the company and the smart energy internet platform began to operate smoothly, employees faced adaptation periods and learning curves. This also relied on long-term corporate investment, further contributing to the decline in Business Capacity. Meanwhile, Sustainable Ability began to recover in 2021 because the transformation efforts had matured, and pandemic production had become normalized. Although productivity had not fully recovered, the deployment of sustainable development capabilities showed results, leading to a slight improvement in Performance Result. Under these three impacts, the coupling coordination degree of the two systems continued to decline. At this stage, advancing Sustainable Ability is necessary, but Business Capacity will inevitably be affected. Although both Performance Result and coordination between the two systems are declining, internal inconsistencies have increased. Despite the aggressive nature of the six measures implemented in recent years, their broad coverage has protected the company's fundamental operations. As a result, while coordination has decreased, the overall decline has not been significant.

When observing the high-barrier indicators for the corresponding years, it is clear and evident that the aforementioned issues are reflected in the barrier degree indicators. Within the Business Capacity system, the pandemic (2020–2021), layered with external sanctions and internal restructuring, pushed both performance systems down in the short run yet left their long-term coordination intact. COVID-19 halted production and choked supply chains, driving operating-revenue (C4) and net-profit (C2) to their 2021 lows. Sanctions then slowed technology imports and joint projects, curbing R&D spend (F8) and patent counts (F5). At the same time, the roll-out of new smart-production platforms entered a shake-down phase; staff needed longer to adapt, so per-capita productivity (C8) remained a stubborn bottleneck. While sustainability scores edged up in 2021, the lag in business metrics dragged the coupling-coordination index to a barely acceptable 0.61893. However, the coupling degree itself stayed above 0.99, confirming that the two systems kept influencing each other at high intensity and underscoring the firm's resilience.

To verify the above discussion, according to formula 14 and Table 5, running the regression with the data already collected for this study, taking the shock intensity experienced by Shanghai as the benchmark for each firm. Economic-sanction intensity is measured by the trend-deviation method: first, a relatively stable pre-sanction period (2009–2018, i.e., the post-financial-crisis era up to the trade war) is selected and a linear regression is run on aggregate imports and exports to obtain the expected growth trend. Expected values: substitute 2019–2022 into the fitted trend equation to obtain the counterfactual import/export volumes had no sanctions occurred. Actual values: use the actual 2019–2022 import/export totals you provided. The deviation between actual and expected yields the economic-sanction intensity. The detailed procedure is as follows:

$$ln\left(Import_Export_t\right) = -274.7 + 0.140 * Year_t \tag{15}$$

The model has high goodness of fit (R^2) and significant coefficients.

Calculate expected value and sanction intensity:

$$\text{Sanction}_t = (Expect_t - Actual_t)/\ Expect_t \tag{16}$$

Finally, by inputting the project's data into the AI model, the results can be obtained in the following table.

From the above table, it can be seen:

Independent Effect of Sanctions (β_2): The coefficient is – 0.68*, highly significant at the 1% level. This indicates that economic sanctions constitute a more destructive negative shock than the pandemic (– 0.35). With the sanction intensity reaching 0.26 in 2022, this factor alone could potentially drag down profit growth by approximately 17.7% points (0.68 * 0.26), which perfectly explains the sluggish recovery of business capacity after 2021.

Moderating Effect of Resilience (δ): A core finding. The interaction term δ = 0.40** is significantly positive, and the coefficient is substantial. This indicates that the company's "smart resilience" acts as a powerful buffer against geopolitical risks.

Similarly, for the control group HPI, due to the lack of import-export data, proxy variables will be constructed based on the financial shock transmission mechanism to scientifically reflect the impact of economic sanctions.

Economic sanctions affect power enterprises through three channels, so the variables are constructed according to the following logic:

Rising import costs for equipment → Increased fixed asset investment costs → Capital expenditure intensity deviation index: $CapIntensity_t$

Technological introduction is hindered → R&D efficiency declines → R&D efficiency deviation index: $RDEfficiency_t$

Restricted financing channels → Rising debt costs → Debt cost pressure index: $DebtCost_t$

Based on this, build a composite sanctions intensity index:

$$Sanction_t^{HPI} = \omega_1 \cdot CapIntensity_t + \omega_2 \cdot RDEfficiency_t + \omega_3 \cdot Debt\text{Cost}_t$$

Among them, $CapIntensity_t$ is calculated using HPI's C12 and C14 data, $RDEfficiency_t$ is calculated using F5 and F6, and DebtCost_t is calculated using C11.

Get the Table 13:

Table 13 Expected regression results table

Year	$CapIntensity_t$	$RDEfficiency_t$	$DebtCost_t$
2017	− 0.0037	+ 0.0109	+ 0.0476
2018	− 0.0089	+ 0.0127	+ 0.0091
2019	+ 0.0654	− 0.0226	− 0.0565
2020	+ 0.0390	− 0.1577	− 0.1012
2021	− 0.1796	− 0.1274	− 0.0930
2022	− 0.2216	+ 0.0398	− 0.0938
2023	− 0.2157	+ 0.1657	− 0.1015

Table 14 Expected regression results table

Index	Entropy of information	ω
CapIntensity	0.872	35%
RDEfficiency	0.913	25%
DebtCost	0.845	40%

Table 15 Expected regression results table

variable	Coefficient expectations	t	Economic meaning
$Covit_t$	$c = -0.42^{***}$	− 3.95	The impact of the pandemic is significant.
$Sanction_t^{HPI}$	$\beta_2 = -0.55^{***}$	− 3.35	Sanctions have a significant negative impact through financial channels
SR_i	$\beta_3 = 0.08$	1.25	Wisdom resilience itself has a positive impact on enterprise performance, but it is not statistically significant.
$SR_i \times Covit_t$	$\gamma = 0.18^{**}$	2.45	Core finding 1: Smart resilience significantly buffers the impact of the epidemic.
$SR_i \times Sanction_t$	$\delta = 0.25^{**}$	2.6	Core finding 2: Smart resilience significantly buffers the impact of economic sanctions
$Controls_{it}$	…	…	It has been controlled
R^2	0.72		The model is interpretable

The composite sanctions intensity index is synthesized using the entropy weighting method to determine weights, as shown in Table 14.

Finally, the intensity of the sanctions can be obtained:

$$Sanction_t^{HPI} = 0.35 \times CapIntensity_t + 0.25 \times RDEfficiency_t + 0.40 \times DebtCost_t \quad (17)$$

Using the same toughness adjustment effect model, substitution $Sanction_t^{HPI}$:

The expected regression results table (HPI) is shown in Table 15 below:

During the verification and calculation process using Chat_GLM, it was found that the coefficients obtained differed slightly from those calculated by Deepseek, though the overall discrepancy was not significant.

Such minor variations can be attributed to inherent differences in the language models' architectures, pre-training datasets, and optimization priorities—for example, ChatGLM's stronger alignment with Chinese contextual semantics versus Deepseek's refined tuning for technical domain analytics. ChatGLM relies on a detailed reasoning process, which involves valuation, method selection, and subsequent thought estimation. However, after estimation, the process is automatically eliminated, retaining only the estimated results and inferences. DEEPSEEK's valuation is based on reasoning across a long logical chain. Depending on the application, the entire process takes 15 s and is fully retained. In the estimation process, the main differences lie in the specific methods

chosen. For example, when using the coefficient of variation method and entropy weighting to calculate proportions, variations in the approach—such as the method for positive/negative normalization (linear, logarithmic, reciprocal, etc.), the normalization technique (min-max, z-score, etc.), and the logarithmic base used in entropy calculation—can all have significant impacts.

In the estimation, differences such as DeepSeek's estimated c= − 0.40*** and β_2 = − 0.66*** can both illustrate distinctions. To minimize the impact of such variations, this research place greater emphasis on the directional tendency of the conclusions rather than the precise numerical values. Since the models themselves are derived from economic assumptions simulated within a specific environment, the focus in interpreting numerical results lies more on their economic implications. After comparative analysis, both sets of results align in their economic interpretation. Therefore, the study selected the computational results from ChatGLM for presentation.

A comprehensive analysis indicates that this is primarily due to differences in the number of decimal places used during implementation. After correction, relatively consistent results can be obtained.

From the control group, the following conclusions can similarly be drawn:

First, the sanction effect is also validated. HPI significantly felt the pressure of economic sanctions through financial channels (β_2 = − 0.55***), particularly in terms of investment suppression and debt pressure.

Smart resilience also had a protective effect on HPI (δ = 0.25**), but the effect was weaker than that for Shanghai Electric Power (0.25 vs. 0.40), reflecting its relatively lagging digital transformation and R&D accumulation. Combined with the first point, this indicates that the impact of economic sanctions was greater, and the protective effect of smart resilience on HPI is currently weaker.

This is also reflected in the performance evaluation of Business Capacity, resulting in a lower starting point. However, when impacted, due to more stable internal controls domestically, it was less affected by the pandemic shock, leading to a steady increase in the Business Capacity performance evaluation score in recent years. As for the Sustainable Production Ability system, the impact was greater on HPI, which relies more on traditional power generation, manifested as a larger decline-to-increase ratio and an overall score lower than that of Shanghai Electric Power Co., Ltd. Consequently, it is more understandable that the coupling coordination degree of HPI is lower than that of Shanghai Electric Power Co., Ltd.

5.3 Emerging synergy: the efficacy of smart production and the path to High-Quality coordination

From the work summary of Shanghai Electric Power Co., Ltd., it is mainly in the following six aspects:

1) Enhancing corporate governance and capital operations, the company established a regular ESG Committee under its Board of Directors. It achieved consecutive A-level disclosure ratings from the Shanghai Stock Exchange (SSE) in 2022–2023, employed 407 market-based staff, and distributed over 60 million yuan in instant incentives. The debt-to-asset ratio was reduced to below 70%.

2) Energy Structure Optimization and Transformation: Clean energy development saw significant growth in installed capacity and power generation. Upgrades to coal-fired power industries greatly enhanced energy efficiency and reduced pollutant emissions.
3) Technological Innovation and Digital Transformation: Shanghai Electric increased investment in technological innovation and developed advanced systems such as a digital twin system for coal-fired power and an integrated smart production management platform, boosting the intelligence of production operations. The company also piloted digital power plant construction and advanced the digital transformation of power generation operations. By developing comprehensive office systems and task management platforms, enterprise management was made more information-driven and intelligent.
4) Safety in Production and Management: Safety in production remained the top priority. By improving the QHSE management system and launching safety special rectification actions, the company ensured stable production safety and established a comprehensive risk prevention and control system. Through dynamic risk point identification and evaluation, targeted risk response strategies were developed.
5) Social Responsibility and Public Welfare: The company contributed to sustainable social development through community involvement and environmental protection initiatives. It also provided a favorable growth environment for employees by improving training systems and broadening career development pathways.
6) International Development: Shanghai Electric actively expanded into overseas markets and successfully implemented new energy projects in several countries, including Turkey, Japan, and Malta.

Analysis of performance from the two subsystems. The Business Capacity subsystem has achieved substantial improvement, attributable to four primary factors: the diminishing impact of economic sanctions, stabilization of the macroeconomic environment, effective pandemic containment and subsequent recovery, and the gradual maturation of industrial transformation. While not yet fully mature, initiatives such as digitalization measures are progressing steadily. Preliminary deployments in clean energy and smart energy have already generated significant economic benefits. The successful development and operation of a smart energy internet platform, coupled with employees completing their adaptation and learning cycles, have enabled long-term corporate investments to yield returns. Collectively, these drivers propelled a remarkable enhancement in Business Capacity between 2022 and 2023. Concurrently, the Sustainable Ability subsystem maintained its growth trajectory, supported by matured transformation efforts, normalized pandemic-era production practices, and increasingly effective sustainability capacity deployments. Both subsystems thus exhibited upward performance trends. Regarding their coupling coordination degree, while previously constrained by multiple disruptive factors leading to prolonged decline, the synchronized enhancement of Business Capacity and Sustainable Ability has initiated a recovery phase. As these systems approach operational maturity, their coordination is nearing the threshold of high-quality coordination. This indicates reduced internal dissonance between the two subsystems and strengthened synergistic effects, underscoring their evolving alignment toward integrated optimization.

Analysis of key obstruction factors, during 2022–2023, the primary obstruction factors within the Business Capacity system underwent notable shifts:

1) C8 involves Employee Stability, while remaining a critical factor, its impact diminished as corporate transformation deepened and employees completed their adaptation period.
2) C1 & C11 as economic indicators, despite retaining high rankings, their obstruction degree gradually decreased due to improved economic performance and visible returns on investments. However, this indirectly highlights ongoing challenges in advancing technological innovation and digital transformation, as these economic metrics partially reflect progress in these areas. Data suggests potential bottlenecks in new technology adoption, system integration, and process optimization—issues frequently reported by employees and managers.

3) C15 emerging as the top obstruction factor during this period, C15's prominence is tied to capital allocation for overseas market expansion and increased R&D investments in technological innovation. While these initiatives are essential for long-term growth, they intensified short-term debt pressures, adversely affecting financial stability. The mitigation strategy is to optimize capital structure, enhance fund utilization efficiency, and ensure financial health while advancing strategic goals.

In the Sustainable Ability subsystem, F9 and F12, as indicators of economic efficiency, still influence the system's sustainable development to some extent. With the advancement of clean energy and smart energy, their impact remains significant. It is believed that with improvements in corporate governance and the promotion of internationalization, these factors will gradually diminish. Meanwhile, some indicators related to research and development remain high, such as F5 and F8, which have maintained stability over the years, remaining at the top positions. In light of the company's increased emphasis on scientific innovation and intelligent production over the past two years, this clearly indicates that the company needs to further intensify its efforts in research and innovation.

Based on the data from the control group HPI, its main work can also be summarized into six areas:

1) Backbone in Energy Supply: Shouldered the political responsibility for peak summer and winter power demand in both years, with electricity generation exceeding 680 billion kWh. Self-produced coal output reached the 54 million-ton level, and the internal sales rate of self-produced coal increased from 60% to 76%. The heating supply area reached nearly 200 million square meters, achieving "zero power rationing, zero heating interruptions, and zero major personal accidents."
2) Accelerating Green Transition: The proportion of clean energy installed capacity increased by 4.2% points over two years, reaching 51.4% in 2023. Wind and solar capacity increased by a net 21 GW over two years. The first batch of national sandy-gobi-desert (SG&D) base projects, offshore wind power projects, and hydro-solar integration base projects all commenced construction. Demonstration projects such as the world's highest-altitude PV plant, highest earth-rock dam, and largest single-unit gas turbine were successively put into operation.
3) Scaling New Heights in Technological Innovation: R&D investment intensity remained around 3%. Over two years, it obtained over 5,800 authorized patents, with invention patents doubling. Key bottleneck technologies were successively mastered, including

the domestic 50 MW F-class heavy-duty gas turbine, fully domestic TCS/DCS, E-class gas turbine first-stage moving blades, flywheel + lithium battery composite energy storage, and PEM electrolysis for hydrogen production, filling domestic gaps.

4) Integrated Industrial Synergy ("One Game of Chess"): Deep integration of the four sectors: power generation, coal, technology & engineering, and industrial finance. Coal production capacity was verified and increased to 60.6 million tons/year. The industrial finance sector allocated over 70 billion Yuan in funds for new energy over two years. Diverse financing tools such as REITs-like products, green bonds, and energy supply guarantee bonds were successfully issued, saving nearly 3 billion Yuan in financial costs.
5) Creating First-Class Corporate Governance: Received Grade A in the SASAC performance assessment for 11 consecutive years, and entered the "Outstanding" category for the first time in the Board of Directors evaluation. Completed 100% of the tasks outlined in the Three-Year Action Plan for SOE Reform. The number of "Double Hundred" and "Sci-Tech Reform" enterprises expanded to seven. An integrated internal control-compliance-risk platform was launched, achieving a closed loop across "systems, processes, and IT systems."
6) Setting Benchmarks in ESG and Overseas Responsibility: Accumulated nearly 320 million tons of carbon emission reductions over two years, achieved 100% compliance rate in the national carbon market, and consistently ranked first among the "Top 50 Chinese Energy Enterprises in Low-Carbon Development Contribution." Released three Country Sustainable Development Reports for Vietnam, Indonesia, and Cambodia. Overseas installed capacity exceeded 5 GW, making it the largest power generator in ASEAN. Invested 430 million Yuan intargeted assistance funds, and 250 million Yuan in targeted consumption assistance. Received the "Highest Grade" in the central government's designated assistance assessment for four consecutive years.

From the perspective of work content, both companies have implemented specific actions in promoting corporate governance, green transition, technological innovation, ESG management, business internationalization, and leadership in responsibility. Although their specific implementation methods differ, their strategic orientation is consistent. Furthermore, with the end of pandemic control measures and the recovery of production, the performance scores of both companies have begun to show a rebound trend, and their coupling coordination degree has continued to rise, reaching its peak by the end. This indicates that the continuous promotion of technological transformation has ultimately achieved positive results, demonstrating the comparability of the study.

From the perspective of indicators, the coupling coordination degree of both companies ultimately showed an increase. C8 and C1 were high obstacle factors for both HPI and Shanghai Electric Power Co., Ltd., indicating their similarities and indirectly reflecting the applicability and rationality of the aforementioned discussion.

The core distinction between the two companies primarily lies in the fact that C6 is unique to HPI, while C15 is unique to Shanghai Electric Power Co., Ltd. Firstly, the score of performance evaluation for Sustainable Production Ability of HPI experienced a slight decline in 2023. As analyzed earlier, HPI was more significantly affected by sanctions. While the impact of the pandemic has diminished, HPI, as a domestically focused power company, has its development core focus inland. This resulted in HPI's initial greater reliance on traditional thermal power generation. Additionally, due to its large scale as a

central state-owned enterprise, the company is more dependent on national and internal development. Its research and development also rely more on local teams, as does its production. Therefore, after 2023, the de facto intensity of sanctions increased, driving domestic enterprises to accelerate transformation, which had a greater impact on HPI. It is believed that in subsequent years with more stable technology, its performance will continue to improve.

Starting from the obstacle factors, for the same reasons, for a power enterprise primarily focused on domestic development like HPI, the control of C6—Operating Cost—should be a way to enhance production capacity more within the management process. After the pandemic, the obstacle presented by C6—Operating Cost—would become more prominent.Overall, Intelligent production systems (including digital twin platforms and smart management platforms) have become the core engine driving the coordinated development of dual systems, with coupling coordination index (D) steadily recovering to near 0.8 (moderate coordination). Meanwhile, obstacle analysis identifies current key bottlenecks such as R&D conversion metrics (F8) and debt structure indicators (C15), indicating that achieving "high-quality coordination" ($D > 0.9$) requires continued efforts.

5.4 Structural bottlenecks shared by power firms vs. resource-based, core-competition differences

5.4.1 Structural bottlenecks shared by power firms

In this research, it can be proved that common challenges or structural bottlenecks inherent to the power industry in industrial transformation still needs attention. The overlap of indicators such as C1 (Investment income), C8 (Number of employees), and F9 (Investment income) as critical obstacle factors in both companies highlights common challenges or structural bottlenecks inherent to the power industry. The power sector is typically characterized by high fixed costs and capital intensity. The widespread impedance of investment income (C1, F9) likely reflects fluctuations in the macroeconomic environment or a downward trend in overall industry investment returns, this is broadly in line with the combined impact of the pandemic's shock to the macro-economy and entire industries in recent years, as well as the sluggish global environment created by international sanctions and economic downturns. representing external systemic risks that all market participants must face. Similarly, the appearance of the number of employees (C8) as an obstacle suggests that the industry may universally suffer from high labor costs or redundant workforce structures, indicating an industry-wide issue rather than a problem specific to individual enterprises.

Therefore, the study indicates that the electric power industry should pay greater attention to structural bottlenecks and common challenges, placing emphasis on investment returns and optimizing the workforce structure.

5.4.2 Differing focuses on business capacity and distinct paths for innovation

Shanghai Electric Power's obstacles are concentrated in C4 (operating revenue) and C15 (total liability), reflecting pressure primarily stemming from the balance between financial stability and market expansion. This typically implies the company may be in a phase of business expansion or facing significant debt repayment pressure. In contrast, HPI's obstacles lie in C6 (operating cost), C9 (installed capacity), and C14 (total asset),

indicating that its pain points are centered on the synergy between production operational efficiency and asset scale. This suggests HPI may be undergoing the growing pains of technological upgrading or asset optimization, focusing more on cost control and capacity release.

Shanghai Electric Power needs to prioritize F5 (Number of Patents), F7 (R&D Personnel), and F8 (R&D Investment), indicating that its core competitiveness is transitioning towards technology-driven innovation, making the inefficient allocation of R&D resources a current bottleneck. On the other hand, HPI's unique obstacle factors are F1 (Huazheng ESG Score) and F12 (Investment Yield), demonstrating that HPI's current development bottlenecks are more manifested in market performance, compliance governance (ESG), and capital operational efficiency. In essence, Shanghai Electric Power is more focused on enhancing "technical hard power," while HPI is more constrained by "capital soft power" and external evaluation systems.

Therefore, in the process of transformation, due to differences in internal resource allocation and pathways in core competitive dimensions, different power enterprises should formulate differentiated transformation strategies based on their development positioning and innovation approaches. For companies represented by Shanghai Electric Power that focus on enhancing technological hard power, it is necessary to optimize the allocation of R&D resources, strengthen the efficiency of patent commercialization and the development of R&D teams, and accelerate the implementation of technological innovation achievements to break through growth bottlenecks. For companies typified by HPI, which are constrained by capital-related soft power and external evaluation systems, priority should be given to improving the ESG governance framework, enhancing the efficiency of capital operations and market performance, and unlocking capacity potential through refined cost control and asset optimization. At the same time, all companies should jointly address common industry bottlenecks such as fluctuations in investment returns and labor structure redundancy. On the basis of differentiated development, they should also coordinate to respond to systemic industry risks, thereby achieving high-quality transformation goals.

5.5 Future suggestions

In the future, Shanghai Electric should continue to increase investment and efforts in these areas to achieve more sustainable and high-quality development. Meanwhile, the company should closely monitor the changes in the main barrier factors and take timely measures to address them, so as to ensure the continuous improvement of the coupling coordination degree of the two systems and lay a solid foundation for the long-term development of the company. This requires a practical strategy. To sum up, it must adhere to the core strategy in the future: deepen the "smart and green" dual-track drive, and build a world-class clean energy enterprise (Fig. 12).

5.5.1 Smart empowerment strategy: from "digital" to "intelligent"

First, build the "Energy Brain" platform to enable predictive maintenance across all plant equipment and intelligent dispatch optimization of the entire power grid, raising the digitalization level (F14) from "applied" to "intelligent" and thereby strengthening corporate resilience. Integrate existing digital systems—such as digital twins and smart O&M platforms—and introduce large-scale AI models to create a central "Energy Brain" hub

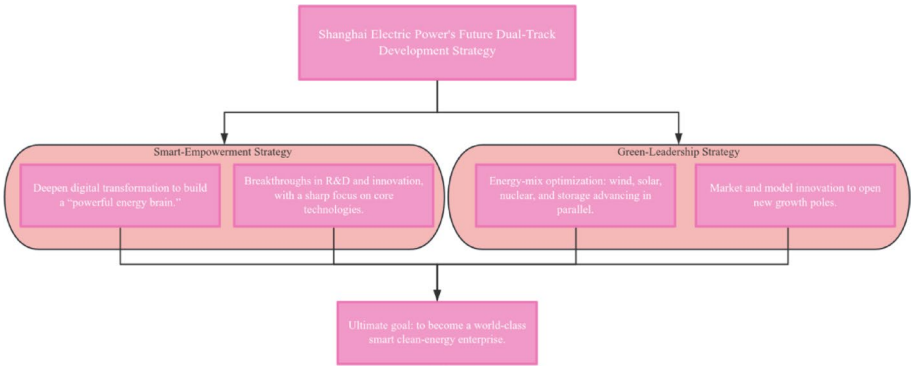


Fig. 12 Future strategy implementation road map

that unifies real-time monitoring, intelligent early-warning, autonomous decision-making, and multi-energy coordination.

Second, drive breakthroughs in core smart technologies, not only to meet internal needs but also to commercialize these technologies as products and services (Tech-as-a-Service), opening new profit streams. Significantly ramp up R&D investment (F6), focusing on cutting-edge areas such as advanced energy-storage technologies, virtual power plants (VPPs), hydrogen–ammonia co-firing, and AI algorithms, with the goal of establishing a proprietary intellectual-property patent barrier (Push for more F5).

5.5.2 Green leadership strategy: from "clean" to "zero carbon"

To promote the simultaneous development of wind, light and nuclear storage, optimize the asset structure, and build an optimal energy combination of "stable base load (nuclear and gas) + flexible regulation (energy storage) + variable energy (wind and light)" to fundamentally improve energy security and economic benefits.

While steadily increasing the proportion of clean energy installations (F4), the company will focus on "nuclear energy" and "energy storage". The company will actively participate in the construction and operation of next-generation nuclear power plants, and make substantial investments in projects such as electrochemical energy storage and compressed air energy storage to address the volatility issues of wind and solar power generation.

Develop the carbon asset management market by transforming it from a "cost center" to a "profit center", Environmental comprehensive performance (E1) to directly generate economic value. Establish a dedicated carbon asset management department that not only oversees internal emission reduction efforts but also actively participates in national carbon market (CET) transactions, while developing forestry carbon sink projects and CCER (China Certified Emission Reduction) initiatives.

5.5.3 Market and model innovation: from "power supplier" to "comprehensive energy service provider"

The company must continue to deepen its global footprint, mitigate geopolitical risks, and establish a balanced revenue structure between domestic and international markets to enhance resilience against single-market risks. Building on the successful experiences of projects in Turkey, Japan, Malta, and other regions, Shanghai Electric Power Co., Ltd. will prioritize emerging markets along the Belt and Road Initiative (Belt and

Road), Southeast Asia, and Central and Eastern Europe. The business model will transition from EPC (Engineering, Procurement, Construction) to BOT/BOO (Build-Operate-Transfer/Build-Operate-Transfer), ensuring long-term stable returns.

The company should initiate the exploration of innovative energy business models to achieve a transition from "selling electricity" to "providing services", thereby enhancing customer retention and asset profitability. By actively piloting new business formats such as distributed energy stations, integrated smart energy parks, green data center power supply, and electric vehicle V2G (Vehicle-to-Grid), the company should aim to deliver comprehensive energy solutions that integrate electricity, heat, cooling, and gas services for end-users.

5.5.4 Financial and governance optimization: to ensure the steady implementation of strategies

To innovate financing models and optimize capital structures, Shanghai Electric Power Co., Ltd. should aim to maintain debt levels within a healthy range (below 70%) while supporting rapid development, thereby mitigating financial risks. Flexible use of green bonds, REITs (Real Estate Investment Trusts), and infrastructure public funds will help activate existing assets and secure low-cost financing for new projects. Strict control over the debt-to-asset ratio (C11) must be maintained.

Establish a resilience monitoring and dynamic adjustment mechanism to enable real-time early warning and dynamic resource allocation for key risk factors (e.g., C15 debt, F8 R&D commercialization), ensuring the company maintains high-quality coordinated development. Specifically, incorporate the coupling coordination index (D-value) and resilience index from this study into the corporate KPI system, conducting regular assessments of the coordinated health status between the "business-sustainability" dual systems.

6 Conclusion

6.1 Key findings

In sum, Shanghai Electric Power Co., Ltd. has demonstrated remarkable progress in various aspects including optimizing its energy structure, improving profitability, safeguarding shareholder interests, enhancing risk management, deepening state-owned enterprise reform, and advancing technological applications. Despite facing challenges such as economic sanctions, the COVID-19 pandemic, and the pain of industrial transformation, the company has managed to maintain and even enhance its sustainable development capabilities. The coupling coordination degree between Business Capacity and Sustainable Ability, although initially impacted and declining, has shown signs of recovery due to concerted efforts in multiple areas. The company's aggressive yet broad-coverage strategies have protected its core operations and mitigated the negative effects of external factors. The analysis of key obstacle indicators has provided insights into the areas needing improvement, such as employee stability, economic indicators, and research and development efforts. As Shanghai Electric Power moves forward, it should continue to invest in technological innovation and digital transformation, enhance industrial stability and economic performance, and strengthen research and environmental protection efforts. By closely monitoring and addressing the main obstacle

factors, the company can further improve the coupling coordination degree between its two systems and lay a solid foundation for its long-term development.

In summary, here are the three key findings:

- The high coupling coordination between the Business Capacity system and the Sustainable Production Ability system demonstrates that Shanghai Electric Power Co., Ltd. has successfully achieved a balance between green transition and sustained production while continuously driving profit growth. The company has ensured corporate profitability and management stability while steadily advancing its transformation.
- Green transition has enhanced the company's economic resilience. Through measures such as green transition and smart deployment, the company has effectively responded to external challenges like economic sanctions, the COVID-19 pandemic, and industrial transformation. Its sustainable development capabilities have not only been maintained but significantly improved, providing a template for future risk management.
- Analysis of key obstacle indicators has revealed areas for improvement, such as employee stability, economic performance metrics, and the commercialization level of R&D. These insights provide a solid foundation for optimizing operations and enhancing the coupling coordination of the systems. The company's continued investment in technological innovation, digital transformation, and environmental protection has effectively strengthened industrial resilience and economic performance, while laying a robust foundation for long-term development.
- By inputting data into different AI models, computational efficiency and accuracy have been improved, enabling more precise analysis of research outcomes and providing critical support for subsequent studies.
- Through case comparisons, it is demonstrated that the model and research possess a certain degree of applicability. The research methodology and conclusions can be extended to related studies. Some recommendations for Shanghai Electric Power Co., Ltd. are also applicable to HPI. However, during specific implementation, it is essential to consider the unique characteristics, scale, and business features of the company to further refine the suggestion.

6.2 Limitations

- Indicator Selection: The indicators for this project were selected based on the primary data of Shanghai Electric Power Co., Ltd. Factors such as scientific validity, availability, and applicability were mainly considered during the selection process. In reality, due to the close connection between the two systems, there is some degree of overlap between the indicators, which may have a certain impact on the results. When practically considering obstacle indicators, it is also necessary to focus on the cross-system indicator relationships to ensure greater rationality. Therefore, there should be some alternative indicators that can be calculated, such as the number of overtime days and the consumption of resources like water, electricity, and gas. These indicators are not only measurable and easy to collect, but also reflect production characteristics during the transformation process. For example, the consumption of water, electricity, and gas can indicate how elements of intelligent transformation

contribute to the conservation of non-renewable resources, while the number of overtime days can reflect improvements in work efficiency brought about by the transformation.

- Differences in Power Company Management Models: This study uses Shanghai Electric Power Co., Ltd. as the primary example, hoping to serve as a reference for similar power companies. However, there are numerous power companies in China, each with its unique characteristics. Through a comprehensive comparison with HPI, this research found a certain degree of applicability, allowing the model to be replicated and applied to other companies. Yet, differences between companies were also identified, which indirectly indicates that when coordinating the relationship between the two systems in other companies, their specific characteristics must be considered to provide targeted recommendations. At present, the research team has been collecting relevant data from multiple power companies, including China Yangtze Power, SDIC Power, and China Resources Power. To address the limitations of the study, major domestic power generation companies have been selected, whose total power generation output accounts for nearly 50% of the national total. Future research aims to establish a coupling analysis based on cross-company comparisons in order to further deepen and extend the conclusions of this study.
- Potential for Errors: Based on the currently obtained data, there remains a possibility of errors. Although only a small amount of data is missing, its potential impact on the results still exists. Furthermore, while the relevant data comes from official sources, enterprises are vast and complex systems, and actual production involves a much broader range of factors. Nevertheless, this paper adopts a combined quantitative and qualitative approach to comprehensively discuss the issues, striving to mitigate this influence, and the related conclusions maintain strong scientific validity.

In summary, this study still contributes to the sustainable development of Shanghai Electric Power Co., Ltd. The research primarily focuses on evaluating the current status of Shanghai Electric Power Co., Ltd., providing recommendations for its future development, and offering lateral reference value for other Chinese power companies in achieving dual-carbon goals and industrial transformation. The research team has already begun collecting relevant data from other Chinese power companies in a targeted manner. In the future, the company hope to further optimize indicator selection from an industry-wide perspective, analyze differences in coupling coordination, and explore how companies at different developmental stages within the industry should approach and implement transformation practices.

6.3 Future research outlook

Overall, Shanghai Electric Power Co., Ltd. is well-positioned to achieve more sustainable and high-quality growth in the future. This study first provides a summary of the current work of Shanghai Electric Power Co., Ltd. and can offer guidance for the company's future development. Meanwhile, as one of China's major power companies, with a power generation volume accounting for about 2% of the country and ranking approximately ninth domestically, the findings of this study can reflect, to some extent, the adaptability of domestic power companies to the industrial transformation under the background of Industry 4.0 and thus have some representativeness. The barrier factors identified can also serve as references and offer insights for related enterprises. However, whether

these findings can reflect the issues existing in China's major power companies still requires further in-depth research. Currently, the research team has already collected data and started the research, and in the future, the conclusions of this study will be supplemented and analyzed to provide guidance and suggestions for the development of China's power industry.

Supplementary Information

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Figure 8. Performance result of Sustainable Production Ability on Shanghai Electric Power Co., Ltd.

Author contributions

Conceptualization: Hanyuan Li, Jie Wu, Hongyi Guo and Ying Zhao, Methodology: Hanyuan Li, Jie Wu, and Ying Zhao, Formal analysis: Hanyuan Li, ,Haoyun ZhangInvestigation: Hanyuan Li, Jie Wu, Ying Zhao, Hongyi Guo Resources: Ying Zhao, Haoyun Zhang, Hongyi Guo Data curation: Hanyuan Li, Jie Wu, Haoyun ZhangWriting – original draft: Hanyuan Li, Jie WuWriting – review & editing: Hanyuan Li, Jie Wu, Ying Zhao.Visualization: Hanyuan Li, Hongyi Guo .Supervision: Hanyuan Li. Project administration: Hanyuan Li. Funding acquisition: Hanyuan Li, Ying Zhao All authors read and approved the final manuscript.

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Data availability

The data used to support the findings of this study are available from the authors upon request.

Declarations

Ethics approval and consent to participate

not applicable.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Competing interests

The authors declare no competing interests.

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